A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 32

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**1 mark each** 1 mark for each of the 2 points below. **1** Most sulfur in coal is present as (solid) iron pyrites (FeS2) **1** Coal is crushed and put in a stream of flowing water: the coal is carried away while the denser pyrites are left behind (washing / streaming) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
01.2	**1 mark each** 1 mark for each of the 3 points below. **1** Flue gases are passed through a bed of crushed calcium carbonate (limestone) **1** Sulfur dioxide reacts (with oxygen and calcium carbonate) to form solid calcium sulfate (and carbon dioxide) **1** If the gases have already been cleaned of smoke, the calcium sulfate is pure enough to be sold as gypsum for building plaster **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **flue gas desulfurisation** → removing sulfur dioxide from flue gases with lime or limestone	3

Do not write outside the box

Question	Marking guidance	Marks
01.3	**1 mark each** 1 mark for each of the 2 points below. **1** The sodium hydrogen sulfite formed can be heated to regenerate sodium sulfite (re-used) and release concentrated sulfur dioxide **1** The pure sulfur dioxide is a valuable raw material for making sulfur / sulfuric acid, so a waste becomes a resource **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **flue gas desulfurisation** → removing sulfur dioxide from flue gases with lime or limestone	2
01.4	**1 mark each** 1 mark for each of the 3 points below. **1 Fluidised bed:** increased surface area allows rapid combustion at lower temperatures so less nitrogen and oxygen react to form NOx (production prevented) **1 Catalytic converter:** platinum / palladium catalyst chemically reduces NOx back to nitrogen and oxygen gases in the exhaust **1** Urea sprayed into the hot flue / exhaust gases reacts with NOx, converting it to nitrogen, carbon dioxide and water **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Catalytic converter** → a catalyst that converts NOx, CO and unburnt hydrocarbons to nitrogen, carbon dioxide and water	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 252-253; AQA spec 3.4.3.2.2-3.4.3.2.3.	
02.1	1 mark for the point below. **1** The time taken for half of a pollutant (released into the environment) to break down / degrade **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **half-life** → the time for the activity, or the amount, to fall by half **Do not credit** • 'after two half-lives nothing is left' - each half-life halves what remains, so some activity always continues	1

Do not write outside the box

Question	Marking guidance	Marks
02.2	**1 mark each** 1 mark for each of the 3 points below. **1** DDT is liposoluble so it is stored in fat / can pass through phospholipid cell membranes and is not excreted (nitrates are water soluble and excreted easily) **1** DDT is persistent (long half-life) so it is not broken down before it is passed to the next trophic level **1** So DDT bioaccumulates in each organism and becomes more concentrated at each successive trophic level **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **persistent** → does not break down / stays in the environment for a long time • **half-life** → the time for the activity, or the amount, to fall by half • **trophic level** → the feeding level in a food chain **Do not credit** • 'after two half-lives nothing is left' - each half-life halves what remains, so some activity always continues • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts	3
02.3	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1** The gas is denser than air so it settled close to the ground / did not rise and disperse **1** Low wind velocity (at night) meant it was not dispersed / diluted **1** Short half-life so it broke down before travelling far **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **half-life** → the time for the activity, or the amount, to fall by half **Do not credit** • 'after two half-lives nothing is left' - each half-life halves what remains, so some activity always continues	2
02.4	**1 mark each** 1 mark for each of the 2 points below. **1** High solubility means nitrates are easily dispersed / leached / carried by water so they affect a larger area (eg rivers, aquifers, the sea) **1** But dispersal dilutes the pollutant so the concentration at any one point may be lower **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **leached** → nutrients or ions washed down out of the soil / washed away by water	2

Do not write outside the box

Question	Marking guidance	Marks
02.5	**Any 2 of 4** 1 mark each for any 2 of the 4 points below, in any order, maximum 2. **1** Noise / sound **1** Heat / thermal pollution **1** Ionising radiation **1** Light **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Ionising radiation** → radiation with enough energy to remove electrons and form ions **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 236-240; AQA spec 3.4.1.	
03.1	**1 mark each** 1 mark for each of the 4 points below. **1 Gravimetry:** gravimeters detect variations in gravity caused by differences in rock density / mass (metal ores usually denser than sedimentary rock) **1** Magnetometry **1 Seismic survey:** sound / shock waves from controlled explosions or a seismic vibrator – echoes recorded by geophones give the depth, density and shape of rock strata **1** Resistivity **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4
03.2	**1 mark each** 1 mark for each of the 3 points below. **1** Trial drilling is the most expensive technique per site **1** Remote sensing (satellite, aircraft, ground surveys) covers large areas at low cost per unit area and identifies promising sites **1** Drilling is then targeted where deposits are likely – it is the only method that produces actual rock samples for chemical analysis to confirm composition and purity (proven reserves) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **remote sensing** → collecting data from aircraft or satellites	3
03.3	**1 mark each** 1 mark for each of the 2 points below. **1** Different minerals emit / reflect infrared at different wavelengths **1** Multispectral sensors detect these wavelengths over the whole surface (eg Landsat 8 every 16 days) giving information on surface mineral composition **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **infrared** → heat radiation / long-wave radiation re-emitted by the Earth	2

Do not write outside the box

Question	Marking guidance	Marks
03.4	1 mark for the point below. **1** Results are available immediately in the field, saving time waiting for laboratory analysis **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 137; AQA spec 3.2.3.4.	
04.1	**1 mark each** 1 mark for each of the 6 points below. **1** Instability / landslips on the hillside, especially after heavy rain - compaction, drainage, establishing vegetation whose roots bind the spoil, regrading slopes **1** Acid drainage from oxidation of pyrites, with toxic metals - collect and treat leachate, add lime to raise pH **1** Spontaneous combustion of residual coal / hydrocarbons - layers of fine-grained material to reduce air flow **1** Lack of nutrients / poor plant growth - add topsoil, sewage sludge, fertiliser **1** Unattractive topography - landscaping / reshaping **1** Dust / heavy metals - cover and keep at high pH **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **leachate** → polluted liquid draining out of waste **Do not credit** • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	6
04.2	**1 mark each** 1 mark for each of the 2 points below. **1** So that plant roots cannot reach the toxic layer **1** Otherwise toxins could be taken up and brought to the surface / dispersed via leaf litter and food chains **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
04.3	**1 mark each** 1 mark for each of the 2 points below. **1** Steel slag is alkaline, producing unusual calcareous / high-pH soil conditions **1** Supports uncommon plants such as cowslips / orchids that cannot compete on fertile neutral soils **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 282-283; AQA spec 3.4.3.2.12.	

Do not write outside the box

Question	Marking guidance	Marks
05.1	**1 mark each** 1 mark for each of the 2 points below. **1** Quadrat size depends on the size / spacing of the organisms: 0.5 m x 0.5 m is typical for ground flora **1** 10 cm x 10 cm is for lichens (too small to contain a representative sample of plants) – 100 m x 100 m is for trees (impractical, too much work) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **quadrat** → a square frame of known area used to sample a fixed area	2
05.2	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** N = 240 – N(N-1) = 240 x 239 = 57 360 **1** Σn(n-1) = 48x47 + 120x119 + 36x35 + 24x23 + 12x11 = 2256 + 14280 + 1260 + 552 + 132 = 18 480 **1** D = 57 360 / 18 480 = 3.10 (accept 3.1) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • **Simpson** → a diversity index that uses both the number of species and their abundance • **Biodiversity** → the variety of life: habitat, species and genetic diversity **Do not credit** • A bare wrong answer with no working. • 'more species always means a higher index' - the evenness of the abundances matters as well	3
05.3	**1 mark each** 1 mark for each of the 2 points below. **1 Lower D means lower biodiversity:** the ungrazed meadow is dominated by one or a few species (eg tall grass) with the others rare (less even) **1** Grazing (plagioclimax management) prevents dominance by vigorous species so abundance is more evenly spread **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodiversity** → the variety of life: habitat, species and genetic diversity • **plagioclimax** → a community held back from its climax by human activity	2

Do not write outside the box

Question	Marking guidance	Marks
05.4	**1 mark each** 1 mark for each of the 2 points below. **1** A frame quadrat flattens tall plants, making percentage cover estimates inaccurate and hiding shorter plants **1** Point quadrat pins are lowered through the vegetation – each touch counts (10 pins = 10% each) without flattening **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **quadrat** → a square frame of known area used to sample a fixed area	2
05.5	1 mark for the point below. **1** Subjective judgement varies between recorders – results are not quantitative so statistical analysis is not possible (but quick for many areas) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 394-395, 401-402; AQA spec 3.7.2.1.	
06.1	**1 mark each** 1 mark for each of the 5 points below. **1** No large dam / reservoir needed so no flooding of farmland, homes or habitats and no barrier across the whole river (run-of-river / micro-hydro diverts part of the flow) **1** Suitable for rivers with a small drop in height and lower flow rates that are common in lowland rural areas **1** Small scale matches local demand – electricity used locally without long transmission lines / grid connection **1** Low construction cost and simple technology maintained by the community **1** Helical turbines tolerate high turbidity / debris without damage and fish pass through unharmed – Kaplan turbines with adjustable blades cope with variable flow **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **HEP** → electricity generated from falling water **Do not credit** • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	5
06.2	1 mark for the point below. **1** Lack of capital / finance, technical knowledge or grid access – seasonal drying of rivers – conflicts with irrigation water – planning / land ownership **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1

Do not write outside the box

Question	Marking guidance	Marks
06.3	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** Output is proportional to flow x efficiency: 100 x 0.10 x (40 / 90) **1** = 4.4 kW (accept 4.4-4.5) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (kW) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2
06.4	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1** Constant regulated flow removes seasonal extremes: dry-season sandbanks used by nesting birds / turtles lost – high flows that clear gravel spawning beds lost – floodplain no longer flooded **1** Sudden flow increases for peak generation raise turbidity, wash away species, reduce light for photosynthesis **1** Sediment trapped so less nutrient replenishment downstream – barrier to migrating fish (salmon, sturgeon) and isolation of populations **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 203-205; AQA spec 3.3.4.1.	
07.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1 C:** Organic (methyl) mercury **Do not credit** • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	1

Do not write outside the box

Question	Marking guidance	Marks
07.2	**1 mark each** 1 mark for each of the 2 points below. **1** Anaerobic microbes converted the inorganic mercury into organic methyl mercury **1** Methyl mercury is liposoluble so it is absorbed, crosses into the brain and biomagnifies along the food chain to fish eaten by people **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	2
07.3	**1 mark each** 1 mark for each of the 2 points below. **1** Most heavy metals are less soluble at high pH **1** So they remain immobile and are not leached into water or absorbed by organisms **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **leached** → nutrients or ions washed down out of the soil / washed away by water **Do not credit** • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 275-279; AQA spec 3.4.3.2.11.	
08.1	**1 mark each** 1 mark for each of the 3 points below. **1** B (about 40:40:20 sand:silt:clay) **1** Balanced mix gives good drainage and aeration from the sand fraction **1** Water retention and nutrient (ion) retention from the clay fraction – easy root penetration and cultivation **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **loam** → a balanced mixture of sand, silt and clay	3

Do not write outside the box

Question	Marking guidance	Marks
08.2	**1 mark each** 1 mark for each of the 3 points below. **1** Large sand particles leave large pore spaces so water drains rapidly (high permeability / aeration) **1** Nutrient ions adsorb onto the (charged) surfaces of clay particles but not onto sand **1** With only 5% clay there is little surface to hold ions so nutrients are leached – low water retention too **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **permeability** → how easily water passes through / lets water through • **adsorb** → nutrient ions sticking to the surfaces of clay and humus • **leached** → nutrients or ions washed down out of the soil / washed away by water	3
08.3	**1 mark each** 1 mark for each of the 2 points below. **1** C (clay) holds much water so has a high thermal capacity: warms and cools slowly – A warms rapidly after cold weather **1** Clay particles held together by capillary water so roots find it hard to push between them – penetration is easy in sandy A **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round.	2
08.4	**1 mark each** 1 mark for each of the 2 points below. **1** Shake a known mass of soil with water in a measuring cylinder and leave to settle – particles settle in order of size (sand first, then silt, then clay; organic matter floats) **1** Measure the depth / volume of each layer to calculate the percentage of each fraction, then use the soil triangle to name the textural class **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 158-160, 412; AQA spec 3.2.5.1.	

Do not write outside the box

Question	Marking guidance	Marks
09.1	**Marks shared** Up to 6 marks in total - share them between the 5 points below according to how completely each one is made. **up to 2 Habitat change:** flooding destroys the previous terrestrial / river habitats but creates a wetland which may be valuable (wetlands are uncommon) **up to 2 Wildlife barrier:** the dam blocks migrating fish (salmon, sturgeon) and prevents recolonisation along the river **up to 2 Changed river regime downstream:** regulated flow removes low-flow periods needed by species such as river turtles nesting on sandbanks and high flows that wash sediment from gravel spawning beds **up to 2 Sedimentation:** sediment settles in the reservoir instead of fertilising the floodplain and building river banks / coastlines downstream, increasing erosion **up to 2 Microclimate:** water's high heat capacity reduces temperature fluctuations – higher wind speeds – more evaporation, humidity, cloud and precipitation downwind **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	6
09.2	**1 mark each** 1 mark for each of the 2 points below. **1** Vegetation / dead organic matter under water decays anaerobically **1** Releasing methane, a powerful greenhouse gas **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	2
09.3	**1 mark each** 1 mark for each of the 2 points below. **1** After heavy rain surplus water is stored, reducing peak flow / flooding downstream **1** In dry periods the dam releases water to maintain river levels / supplies **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 117; AQA spec 3.2.2.1.	

Do not write outside the box

Question	Marking guidance	Marks
10.1	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • Organic nutrients (sewage, manure, silage effluent, paper mill / tannery / food processing effluent) are food for aerobic microbes whose respiration deoxygenates the water, killing fish and invertebrates – bacterial growths cover the bed, blocking light • Pathogens in sewage (cholera, typhoid, dysentery) spread via contaminated water – decay of organic matter releases inorganic nutrients • **Inorganic nutrients:** nitrates (fertiliser / manure runoff, leaching) and phosphates (sewage effluent, detergents, eroded soil) cause cultural eutrophication: algal blooms, shading of macrophytes, cyanobacterial toxins, night-time and end-of-summer deoxygenation • **Human health:** nitrites and methaemoglobinaemia – possible carcinogen via nitrosamines • **Control of organic wastes:** sewage treatment (pre-treatment, primary sedimentation removing >95% organic matter, aeration / activated sludge or trickling filters, anaerobic digestion of sludge with biogas) • **Control of nutrients:** tertiary phosphate stripping with iron(III) sulfate, denitrification in filter beds, micro-strainers / UV – Nitrate Vulnerable Zones, fertiliser timing / amount, low-solubility fertilisers, legumes, manure management – dredging sediments • **Evaluation:** sewage treatment has virtually eliminated deoxygenation and waterborne disease in developed countries but storm overflows and combined sewers still discharge untreated sewage – tertiary treatment is expensive so only used at sensitive sites – diffuse agricultural nitrate is hard to control and aquifers respond slowly – phosphate legacy in sediments • Judgement on point-source (sewage) control being more successful than diffuse-source (agriculture) control	9

Do not write outside the box

Question	Marking guidance	Marks
	Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content. • **leaching** → nutrients or ions washed down out of the soil / washed away by water • **eutrophication** → nutrient enrichment of water leading to algal growth • **deoxygenation** → loss of dissolved oxygen from the water **Do not credit** • 'the algae use up all the oxygen' on its own - the oxygen is used by aerobic decomposers breaking down the dead algae	
	Model answer - Level 3 (9 marks)	
	Organic wastes - sewage, manure, silage effluent, paper-mill and food-processing effluent - are food for aerobic bacteria, whose respiration strips dissolved oxygen from the water so fish and invertebrates suffocate and bacterial growths smother the bed; sewage also carries pathogens such as cholera and typhoid. Inorganic nutrients act differently: nitrate leached from fertiliser and manure and phosphate from sewage, detergents and eroded soil cause cultural eutrophication, in which algal blooms shade macrophytes, cyanobacteria release toxins and night-time respiration and decay deoxygenate the water. Nitrate in drinking water can cause methaemoglobinaemia and may form carcinogenic nitrosamines.	
	Point sources are controlled by sewage treatment: screening and grit removal, primary sedimentation removing most organic solids, aeration with activated sludge or trickling filters, and anaerobic digestion of sludge yielding biogas. Nutrients need tertiary treatment - phosphate stripping with iron(III) sulfate, denitrifying filter beds, micro-strainers and UV. Diffuse agricultural sources are managed by Nitrate Vulnerable Zones, limits on the timing and amount of fertiliser, low-solubility fertilisers, legumes, manure management and buffer strips.	
	Sewage treatment has virtually eliminated river deoxygenation and waterborne disease in developed countries, but storm overflows on combined sewers still discharge untreated sewage, and tertiary treatment is expensive so it is used only where eutrophication is a risk. Diffuse nitrate from farmland is much harder to control because it comes from thousands of sources and depends on farmers' behaviour and the weather. Control of point sources has therefore been far more successful than control of diffuse ones.	
	Weaker answer - Level 1 (3 marks)	
	Sewage in rivers uses up the oxygen because bacteria break it down, so fish die. Fertilisers wash into rivers and cause eutrophication, where algae grow and block the light and then the water runs out of oxygen. Sewage is treated in sewage works to clean it before it goes into rivers. Farmers should use less fertiliser. Sewage treatment works well but fertiliser is still a problem.	
	What makes the difference	
	• Level 1 has the outline of both mechanisms but Level 3 explains them precisely (aerobic respiration, shading, night-time respiration, pathogens, health effects) and names the pollutant sources.	
	• Level 3 describes the stages of sewage treatment and the specific nutrient controls; Level 1 says 'sewage works' and 'less fertiliser'.	

Do not write outside the box

Question	Marking guidance	Marks
	• Level 3's evaluation gives reasons (storm overflows, cost of tertiary treatment, diffuse sources) for its point-vs-diffuse judgement; Level 1 asserts it.	
10.2	1 mark for the point below. **1** Algae / cyanobacteria absorb nutrients directly from the water (not the sediments) so they use dissolved nutrients more easily and grow rapidly **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **eutrophication** → nutrient enrichment of water leading to algal growth **Do not credit** • 'the algae use up all the oxygen' on its own - the oxygen is used by aerobic decomposers breaking down the dead algae	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 266-272; AQA spec 3.4.3.2.9.	

Do not write outside the box

Question	Marking guidance	Marks
11.1	**Soil is often described as a non-renewable resource. Discuss the causes and consequences of soil degradation and evaluate the strategies that can be used to manage soils sustainably.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Features of fertile soil:** water, nutrients, air, organic matter, pH, biota, texture, structure, depth – slow rate of soil formation vs rapid erosion (dynamic equilibrium broken) • **Causes of degradation:** vegetation removal, ploughing vulnerable soils, overgrazing, reduced organic matter and biota, cultivating steep slopes, compaction, salinisation from irrigation, nutrient depletion, pollution • **Types of erosion:** rain splash, runoff, wind, slumping – consequences: reduced productivity, river and reservoir sedimentation, flooding, coastal sedimentation, particulates, desertification, landslides • USLE as a tool for estimating loss and testing management • **Management:** long-term crops, low / zero tillage, contour ploughing, tied ridging, terracing, stone rows, windbreaks, multicropping / strip cropping, mulching, organic matter, livestock control, drainage / pH / nutrient management, conservation of soil biota • Links to sustainable agriculture, carbon storage, hydrological cycle, circular economy	25
	• Evaluation and judgement on which strategies are most effective in different environments *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 158-169, 328-337; AQA spec 3.2.5, 3.5.1.3, 3.5.1.5.	

Do not write outside the box

Question	Marking guidance	Marks
11.2	**Explain how feedback mechanisms and tipping points influence the rate of global climate change and discuss the implications for the strategies needed to control it.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Negative feedback:** increased low-level cloud (albedo), increased photosynthesis and sequestration – hydrological cycle evaporation-precipitation • **Positive feedback:** melting permafrost (methane), reduced albedo of snow / ice / sea ice, methane hydrate release, ocean acidification and coral loss, forest and peat fires, soil decomposition, water vapour, cirrus clouds • **Tipping points:** self-sustaining processes – why control must precede them – examples • **Implications:** urgency and precautionary principle – need for deep cuts in greenhouse gases (energy, transport, buildings, agriculture, forestry, CCS) rather than only adaptation – geoengineering risks • Difficulty of prediction because feedbacks interact at different scales – models – international agreements and per capita footprints • Judgement on whether current strategies are adequate	25
	Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion. **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 103-107, 371-372; AQA spec 3.2.1.2, 3.6.1.	

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